

ActInf Livestream #025.0 ~ "The Computational Boundary of a Self"

Livestream | 1:40:19

foreign

go for it hello everyone welcome to

acton flab live stream number 25.0

today is july 1st 2021 and we're going

to be talking about this paper

the computational boundary of a self

developmental bioelectricity

drives multicellularity and skill free

cognition

i'm blue and i'm here with daniel

awesome so welcome to

the active inference lab everyone we are

a participatory online lab that is

communicating

learning and practicing applied active

inference

you can find us at the links here on

this page this is a recorded

and archived live stream so please

provide

us with feedback so that we can improve

our work all backgrounds and

perspectives are welcome here

we will be following good video

etiquette for live streams

here at the short link you'll find all

of the live streams and different series

that we do

in the communications unit of the active

inference lab

and today we're doing

the contextualizing paper the dot zero

video for two

upcoming discussions in the first half

of july 2021

on the 6th and the 13th when we have discussions

25.1 and 25.2 on this paper and hopefully the author joining today in active lab live stream number 25.0

we're going to be trying to set some context and give an introduction to the following paper the computational boundary of a self

developmental bioelectricity drives multicellularity

and scale free cognition by mike levin

and this video is just an introduction

to some of the ideas it's not a review

or a final word it's kind of like a

three-way intersection where we have people

who might be within the active inference community looking to be exposed to some different areas like developmental biology or cognitive science

and then another group of people who are coming from the developmental biology or cognitive science background

and curious about active inference and

then also we hope that this is exciting

and interesting even for people who are not familiar

with active inference or developmental biology or cognitive science

we will hopefully try to connect it

with some broader questions

so we're going to walk through some of

the aims and claims of the paper the

abstract and the roadmap

and bring a few big questions

and some key points of the paper so that

whether you read the paper or not you'll hopefully be in a good spot to ask questions and learn more and of course in the 1 and 2 videos in the coming weeks we'll be discussing the same paper so save and submit your questions and let us know if you'd like to participate or contribute in any way so here we are on the paper itself that's a screenshot of the cover and the aims and claims and i'll read them so that then daniel you can give a first thought on what you thought was cool or what you thought was important about the paper so the aims and claims of the paper first they define individuals themselves in a way that facilitates taxonomy comparison and communication with evolved created biological artificial and exobiological agents he says i propose a fundamental definition of an individual based on the ability to pursue goals at an appropriate level of scale and organization and suggest a formalism for defining and comparing the cognitive capacities of highly diverse types of agents second he proposes a plausible naturalistic framework for the evolutionary scale-up of cognition from the earliest origins of life hypothesizing about the forces that drove it and the major transitions along

the continuum

and the goal of this research program is
to show how complex agency and goal
directiveness

evolves naturally from ancient
mechanisms so

daniel what do you think about that
nice work well i thought the paper was
pretty

exciting because it touches on just so
many big topics like

goal setting and agency cognition in
natural and artificial systems

multi-scale systems and then it links it
in a way that people might not
expect which is through developmental
biology

and bioelectricity which are kind of
like processes that these life forms or
other forms engage in so to be able to
connect

things that are awesome about systems
that matter like again

their goal seeking nature their ability
to do cognition and then connect that to
a process theory

it's kind of what we're all about in
active inference connecting

the cool features of the world to
process theories that are tractable
cool so moving on to the abstract i'll

read the first part and then you can
read the next slide

so all epistemic agents physically
consist of parts that must somehow
comprise an integrated cognitive self
biological individuals consist of
subunits organs cells and molecular

networks that are themselves complex
and competent in their own native
contexts how to how do coherent
biological individuals result from the
activity of smaller sub-agents to
understand the evolution
and function of metazoan creatures
bodies and minds it is essential to
conceptually explore the origin of
multicellularity
and the scaling of basal cognition of
individual cells
into a coherent larger organism
so i find this line of research truly
fascinating and really a burning
question for me
is how information is consolidated or
prioritized across skills
whether it's core screening or salience
or
something else so do you want to read
the next part
yep just one thought on that and it's
going to be a theme that's returned to
it's that cognition or agency doesn't
just
blink into existence at a larger level
of analysis
it's actually composed of smaller
cognitive units
and so it's kind of like cognition all
the way down that helps us
integrate across scales but also
recognize what's special about larger
levels of organization like culture
so part two of the abstract in this
article
i synthesize ideas in cognitive science

evolutionary biology and developmental
physiology
towards a hypothesis about the origin of
individuality
scale-free cognition i propose a
fundamental definition
of an individual based upon the ability
to pursue goals at an appropriate level
of scale and organization
and suggest a formalism for defining and
comparing
the cognitive capacities of highly
diverse types of agents
any self is demarcated by a
computational surface
the spatial temporal boundary of events
that it can measure
model and try to affect the surface sets
a functional boundary
a cognitive light cone which defines the
scale
and limits of its cognition i
hypothesize that
higher level goal-directed activity and
agency
resulting in larger cognitive boundaries
evolves from the primal homeostatic
drive
of living things to reduce stress the
difference between current conditions
and life optimal conditions
so i like this goal directed like aspect
of this and it really kind of implies to
me that the
it's like biological cognition has to be
biological but i wonder about like
you know what is the goal of of
molecules in their like organization

right is that like a goal directed
activity like the protons and neutrons
and electrons
want to like reduce their uncertainty
about where they are in space or how
does it work at like a non-biological
level makes me wonder that
or strain strain and stress those are
things we psychologically experience but
then you can have you know like a
ring with four carbons it's under strain
and the carbons
they seek to do what well get into a
less stressed position
like a unstrained arrangement for sure
the next part the mechanisms of
developmental bioelectricity
the ability of all cells to form
electrical networks that process
information
suggests a plausible set of gradual
evolutionary steps that naturally lead
from
physiological homeostasis in single
cells
to memory prediction and ultimately
complex cognitive agents
via scale up of the basic drive of
infotaxis
recent data on the molecular mechanisms
of pre-neural bioelectricity suggest a
model of how
increasingly sophisticated cognitive
functions emerge smoothly from cell to
cell communication
used to guide embryogenesis and
regeneration
this set of hypotheses provides a novel

perspective
on numerous phenomena such as cancer and
makes several unique
testable predictions for
interdisciplinary research that have
implications not only for evolutionary
developmental biology
but also for biomedicine and perhaps
artificial intelligence
and exobiology do you have any thoughts
on that daniel
i guess it's just cool to see
development and cognition
come on to equal grounding because there
was ecology evolution and development
eco evo devo
and then it's sort of that's natural
things happening and then
often cognition and especially
computational perspectives on cognition
are seen as unnatural or designed and so
again
all of these pieces are going to be
woven together with a few
pieces like bioelectricity and light
cones that maybe people
aren't uh expecting cool
so the road map of the paper we're gonna
go through the introduction
then the paper is broken up into lots of
different parts what is the self how do
you define an individual
body patterning cognition
multicellularity versus cancer
defining individuation from a cognitive
perspective
the agent's evolutionary backstory
conclusion future outlook discussion and

then uh some predictions and research
program

and then in the end uh what does it feel
like to be a pancreas which

i i renamed that part but maybe you
changed it back so we'll see

do you want to read over the keywords
so here's the provided keywords were
development

bioelectricity and the related gap
junctions

primitive cognition and our favorite
active inference

so development i like this quote from
smith seth

mary and maynard smith which i can never
say or says last name i feel bad sorry
yours

developmental biology can be seen as the
study of how information
in the genome is translated into the
adult structure

and evolutionary biology of how the
information came to be there

in the first place so that's a kind of
cool definition of development

do you have any thoughts on that
it reminds me of earlier discussions we
had about

tin bergens four y's which are kind of
related to aristotle's four whys it's

like why is there a foot

well the easy sort of material answer is
because there's matter in the shape of a
foot

but maybe you want to know something
that's historical and over really long
periods of time

it's a story about evolution how the
foot came to be selected on through time
but development is just right there in
the middle where it's
the story for that individual how that
foot came to be

um thanks said blue could you turn down
my mic

it um apparently it's a little loud or
distorted thank you for the feedback
yeah if anybody has comments on like the
audio

um blue's doing an awesome job of first
broadcaster role

so thanks ed cool all right hopefully
that's a little bit better

all right uh

bioelectricity so this is a quote from
the paper

developmental bioelectricity is the
ubiquitous exchange of slowly changing
ion-based voltage signals within and
among cells

all cells are electrically active and
modern neurons evolved from pre-neural
precursors that were already reaping the
benefits of ionic signaling for
computation

computation ionic signaling for
computation

so there's some ion channels i like that
cool like digital

um image of ion channels and cells and
of course this painting by alex gray
like isn't officially

bioelectricity but whenever i think of
the word bioelectricity i always think
of this picture

do you have any thoughts and to connect
the developmental aspect to what is
mentioned
here about the cells being electrically
active
the neural tissue arises during
development from neuro ectoderm
so before there was sort of an inner
layer of these electrically specialized
cells
ones that we call neurons there
was an outer layer and we see that
recapitulated
in embryogenesis when the neural tube is
closed
so that's combining sort of ontogeny
recapitulates phylogeny which is just to
say that
the development of the organism carries
some
resonance with the way that the organism
evolved through time
and then just to remind that every cell
has
those pumps that maintain charge maybe
not all of them are involved
in rapid fire signaling but to maintain
ionic gradients
and therefore bioelectric gradients is
essentially
what cells do with remembrance cool
and awesome segway into talking about
gap junctions
so this is like something that's near
and dear to my heart with my like
neuroscience background
so gap junctions like were originally
attributed to like

neurons right like that's it like gap
junctions are there at the synapse and
you know they facilitate the electrical
signaling between neural cells but
but really they're um channels that
permit cell to cell
transfer of ions directly and they were
originally described in nerve and muscle
but they're really found in virtually
all cells that are present
in solid tissues um and then i just did
a quick like
brief look up of a recent review of gap
junctions and it's cool like they're
involved in all kinds of
um things right like so self
proliferation immune response migration
apoptosis carcinogenesis
hyperproliferative skin disorders
lymphatic vessel diseases
inflammatory lung diseases that's like
asthma liver injury and neoplastic
disorders
that's also cancer so i thought that
that was kind of a neat
um thing any thoughts on this one daniel
just that active inference helps us
think about how
agents communicate and interact so if
there's two cells and they have totally
disjoint membranes
then they can both modify a niche in
common so they could pump ions in and
out of the fluid
like the extracellular matrix that's
surrounding the cells
um so there's a common niche that
they're sharing that's like the least

connected they could be
and then there are also ways in which
they can have a long-term connection
with a higher throughput so we're seeing
a lot of these ideas of communication
where you can talk about these two
agents now directly exchanging
information
with each other as their electrochemical
niche
rather than just jointly modifying the
shared fluid that they're in
cool okay next
all right primitive cognition see if i
can play this video did you want me to
play it
sure if it works on your side let's see
it
perfect
role of actin in cell motility stunning
3d animation
maybe we don't need the sound but um
next well what um is written in the
paper
is the emerging field known as basal
cognition
tracks the evolutionary history of
learning and decision making processes
beginning from the dynamic
problem-solving capacities of cellular
and subcellular forms many examples of
memory anticipation context-dependent
decision-making and learning
are exhibited by organisms from yeast
and bacteria
to plants and somatic cells
cool so down at the bottom what does
cognition or intelligent behavior look

like across scales
where and how does cognition arise and
how does it develop or evolve convergent
and divergent features
and where does active inference come
into play
i think that first question is fun
because it's a little
easy to off the cuff just say well i i
don't recognize the intelligence of a
cell
because it's not um you know doing an
undergrad course
or it's not doing skillful performance
like we talked about in a
previous discussion so will we have a
human-centric
conception of intelligence tied up with
all of our cultural understandings
or is there something about intelligent
behavior that might help us
recognize it across scales and that's
what turns us to these fundamentals
like memory anticipation context
dependent decision making
and then once we open up to that
functional level
of intelligent behavior in cognition we
start seeing it everywhere
and so instead of just seeing these
cells and their behavior as
taking part of an intelligent system
that again only blinks into existence at
the top level
they're in and of themselves intelligent
actors
awesome so let's see
the video got me locked in here sorry

how about that you want to take us
through that slide daniel
yep so it's written in the paper
i propose oh well we'll come back to um
homeostasis in a second at the top it's
written
developmental bioelectricity is the
ubiquitous exchange
of slowly changing ion-based voltage
signals within and among cells
all cells are electrode electrically
active and we talked about that just a
few slides ago
how ancient are these mechanisms how
early was bioelectric coordination
exploited by evolution so this is
raising the notion
that early cognition was biomechanical
it was biophysical it involved
morphological
computation like we saw on the previous
slide with the movie
of cells but also electrical affordances
existed from the beginning and so if
biophysical affordances existed from the
beginning
and allowed for cognition to arise then
it makes just as much sense to say that
electrical affordances
for cells to communicate and do these
types of cognitive processes
also were exploited very early in
evolution
so primitive meaning at a smaller level
but also
earlier on and suggesting that it's
bioelectric was one of the earlier uh
mechanisms of cognition is pretty cool

and i like this quote from the paper
that says the ability to operate toward
a region and state space may be the
primitive origin
of complex cognitive systems that can
entertain counter factials
which is to remember or anticipate
events that are not occurring
right now so we'll touch back on
homeostasis but how are the goals
related to homeostasis
how does bioelectricity play a role in
homeostasis and i think we come back
around to that
on 17 i wanted to give an example about
where bioelectricity and development
came together with cognition and
for some people genetic reductionism
makes it so that you know the story is
not complete until we know
what gene is involved and i think that
they'd find this
recent 2021 paper from drosophila fruit
flies to be really interesting
because what they do with a series of
experiments is actually demonstrate the
importance of
membrane depolarization which is to say
bioelectric
capacity in uh
gene regulatory networks so like hh and
ptc
are uh proteins that are produced by
loci and dna and they're part of
feedback loops
involving the depolarization of
membranes so they show that
regulation of that membrane potential

has an important role
in signaling hedgehog and that there's a
mutually reinforcing relationship
between these two processes and then
what does that allow
well it allows morphological computation
of this developing disk of
the fly that later turns into the wing
and so there's advantages and
disadvantages like
with diffusible molecules there's a
tremendous amount of expressivity
with what that molecule is like you have
an infinite number of hormone
possibilities
but on the other hand it's very slowly
diffusing and
maybe you can only signal a short range
on the other hand there's only like
one electricity there's not like the
blue membrane potential and you know the
pink membrane potential
as parallel membrane potential there's
only one membrane potential
so it's kind of a big knob to turn
however
as shown here it really does play a
functional role
in stabilizing gene regulatory networks
if you were missing
that part of the puzzle it would be as
incomplete as if you were missing a
critical signaling molecule
so i suspect that as advances the
kind that are going to be referenced in
this paper uh advances
in being able to visualize and track and
manipulate bioelectric fields

become more accessible that more and more systems that were thought to be purely chemical or purely mechanical i think that there'll be a more important role for bioelectricity right it is like one big surface one membrane potential so you have one membrane and that's like the the differentiator right between that's the boundary right is what is enclosed or encapsulated in the membrane versus what is not

uh okay so here we are active inference this is the final keyword and i pulled this image out from the active inference lab

um a previous uh live stream that we did with um ines and uh tomas or thomas so this is the free energy principle computation and computationalism and realism and tragedy and so this is just the basic action perception loop so you have active states external states sensory states and internal states and so the both of the active states and the sensory states are the blanket states right so you have the the internal can act on the external uh via these two blanket states so this is all of the sensory perception of the organism and the degree on which they're able to act

on the environment do you want to add
any thoughts here daniel

it's just that active inference is a
unifying framework for thinking about
perception cognition and action and it
does that in
a scale-free way that's grounded in
bayesian statistics and physics
so it's going to be deployed as a way to
help us
look for different kinds of behavior
across scales because the internal state
could be internal to a cell
it could be internal to a tissue
internal to the organism etc
and it's that expressivity of the active
inference framework
that is going to come into play in the
paper

awesome and it's a process a scale-free
process

okay active inference as brought up in
this paper

the author says agents scaled up by
evolving from
basic homeostatic loops driven by active
inference
surprise minimization via addition of
delays which are memory
anticipation which is inference and
networks

spatially distributed processing that
enables learning and progressive
abstraction or generalization from the
data
gathering into larger collectives with
optimal information structure also
improves the computational ie predictive

capabilities
and gives rise to functional
relationships such as memories encoded
goal states
test operate test exit loops that exist
over and above
any individual member these levels
coexist enabling numerous coherent
selves of different skills to be
implemented by any collection of living
matter
that's intense uh the other one of the
other pieces about active
inference that differentiates it from a
framework like reinforcement learning
is right there um that there's the
minimization of surprise and in order to
minimize surprise through time you kind
of have to be
doing two things that at first pass
don't seem like they're going to
facilitate each other which is the
pursuit of
what you already know is going to be
useful that's called pragmatic value
as well as the pursuit of sampling
things that
will be informative that's epidemic
value and so
in active inference the way that agents
through deep time
minimize their surprise about perception
that's what's referenced in the top
paragraph about perceptual control
control theory
the way that the agents minimize their
surprise about their future
expectatio expectations of perception

is going to be by jointly working on the
pragmatic and
epidemic aspects of their behavior
and thinking about what behavior is
going to be minimizing free energy in
terms of
kind of existing on the optimal
trade-off where you're
performing but also learning so these
are a few features of active inference
that differentiate it and then if we
think about like a growth cone of a
neuron
is it doing reinforcement learning is it
doing reward maximization
it just barely makes sense but once we
start to think about surprise
minimization
like it's endowed with an evolutionary
prior it expects to connect to a
neuromuscular junction
then maybe some of its searching
behavior makes more sense
and in the top section also the author
you know states that the sensory
input layer is what enables active
inference to operate
so because there's receptors and
channels that that's the perception the
perception
that for a cell um yeah anyway just
thought i'd add that in
okay greeting sarah
hi sarah uh
daniel do you want to read these uh core
assumptions and i'll fix the thing
thank you classic so i'm only gonna
read the uh the two

uh highlighted parts here because these are paragraphs that people want more detail on they can definitely read the paper to learn more about there's three core assumptions that are laid out really clearly so i appreciated that about this paper that it makes clear assumptions and shows how those assumptions come into play and then basically there's two avenues for disagreeing i guess one would be i think those are invalid assumptions but they're um properly applied or one can think that there are valid assumptions but improperly applied and so that helps structure discussion with the contents of the paper so the first core assumption is a commitment to evolution every capacity has a natural history and emerged from simpler variants um i guess also evolution can perceive through a reduction of complexity so eventually it had to have emerged from simpler variants but in the short time i guess it could have emerged from more complex variants but the commitment to evolution means the commitment um to the dispelling of magical thinking and not looking for some sort of a metaphysical answer to our physical questions and the second assumption of the three

is it is assumed that all metaphors are
to be judged
by their utility in driving scientific
progress
and there is not a binary categorization
of scientific pictures which should be
taken
literally or not which can be decided a
priori so we're not going around with um
post-it notes saying what's real or
what's not
we're using utility as a guiding metric
um we can actually just take a quick
visualization and then sarah you can
say hello and give any uh thoughts if
you'd like
but we previously in live stream 15 and
20
talked a lot about this continuum from
realism to instrumentalism
like is active inference the territory
itself
that's realism we're really describing
what's really there when we make an
active inference model
or instrumentalism on the right side
here
is like active inference as a map we're
just using it
as something that could be um used to
capture some features of a system
and then used instrumentally like a
model and then
here we've introduced sort of this third
vector of utility
which is like okay whether it's the map
or the territory orthogonal to that
is is it useful and so this is a space

that's describing not just where we can think about active inference existing but also ways to evaluate different models not that one is better than the other but these are just different dimensions that models exist on all right okay maybe go to the third core assumption on to the third one so much of the discussion centers around goals related to teleology a hotly debated topic here goal directedness is taken in the non-magical cybernetic engineering and control theory sense of a feedback system that operates to maximize some specific state of affairs which can be modeled as a dynamical system with the tractors in its state space so to reiterate these three assumptions the first one is evolution the second one is utility and the third one is a non-magical sense of goal directedness so these three assumptions play nice with each other and then they also lead directly to this final statement that except for a few brief remarks at the end no claims about consciousness defined rapidly as first person experience or sense of self as qualia are made all the examples concern

functional
third person objective capacities
computation and behaviors
so we're kind of like outside of systems
looking at them
as an investigator this isn't about the
first person perspective
and there's always so much fascination
and interest
in what consciousness is but in a way
these three assumptions
go a really long distance towards just
saying hey let's look at the
systems behavior as we observe it rather
than as we imagined it to be
like so that's at least the starting
point for
a great discussion um
before we jump into the figures sarah do
you want to
give any thoughts or say hello
uh sure yeah i am i
i'm kind of coming in pretty disoriented
but uh
tried to follow michael's work uh quite
a bit and i'm enjoying
uh discussing it from whatever angle
comes up
and oh i don't know uh background who
cares um
i'm in philosophy right now um yeah
that's fine
nice well any slide you want to
chime in on is great so
find the slides okay they're yeah anyway
um
um go for figure one blue if you want
sure so this is a continuum of

cognitive powers so and this is just the first part of the figure we'll show the next part in the next slide so this is a

tote

which is test operate test exit loop

uh which is a schema of a basic homeostatic cycle so the author says that by continuously taking action to minimize the distance the difference in error between the current state of affairs and a set point describing a different possible future state of affairs it enables a system to pursue goals despite perturbations from the outside world and intrinsic noise

so this is i thought cool like you know the first like possible counterfactual like yes it's 75 degrees and i have plenty of sodium and potassium and you know calcium but but what if i don't always what if it's not always this way right what if what if it my state changes so like that the first counterfactual maybe enabled this kind of homeostatic feedback loop i thought that that was a neat idea put forward in the paper so how is this related to active inference or systems engineering daniel well the terminology reminded me a lot more of engineering than we often see like sensors and actuators

it makes it sound like it's going to be
a robot
and also this like test exit sounds like
we're going to be doing sort of software
evaluation
but we can pretty clearly see the
skeleton of
action perception loops whether it's
framed as it was
a couple slides ago with internal
external states
and then blanket states or here where
you have to basically have the
difference between the desired and
expected value
leading to policy selection not shown
here leading to action
actions cause events in the environment
on the yonder side of the blanket
and then the state of the external
environment
influences the measurements on the
inbound side which
are again contrasted with the idealized
state
and that leads to another round of
action selection so this is a
engineering way of laying out the test
operate test exit loop
but who knows we're able to jump from
active inference pretty
nicely into a more engineering frame and
so
perhaps engineering could be integrated
closely into active inference
yeah it seems like the blanket states
like the sensors and actuators like
our um you know the action perception

part are here in the middle and
so it kind of reminds me of the active
inference loop like diagram that we
showed earlier so it's interesting
uh next part do you want to go through
this one danielle
yeah so in figure 1 b the
caption reads that the scale is showing
how different types of activity and
systems can be ranked
according to their degree of
purpose-ness so again
think about purpose in the third person
not the first person
it's not the sense of purpose but more
um
what i believe hofstetter called the
anti-species
i'm not sure to pronounce it but it was
like think about the most mindless
behavior
and then push away from that and that's
what you can call purposeful
rather than imagining that there's like
this pure crystallized
purposeful behavior and it sort of goes
from
bottom to top in terms of the degree of
purpose
ability and on the bottom we see
um basically just action that doesn't
appear
to have any kind of um
sense or purpose to it like maybe gas
molecules bumping around
and then we progress through higher and
higher levels of
feedback loops and then each one of

these brackets it's kind of like a distinction
so like starting left behavior you can take the right branch or the bottom branch passive behavior like a ball that's being thrown or active behavior
okay active behavior is it gonna be non-random
um or random like many things are random but you know is it a wind-up toy that's just doing something that is not not appearing to be too purposeful and you continue up and up the branch choosing the higher branch each time until you get to these first and second order predictions and higher order recursive logics which are seen as higher and higher levels of purpose
cool do you have any thoughts sarah
okay uh daniel do you want to take this one
what is an individual well there's so many ways to go about this and as an ant researcher that's always the debate like okay is the individual the six-legged next mate and then that makes the colony like a society or a super individual or is the colony the individual and then the nest mate is like the tissue so there's a lot of discussion around what constitutes an individual and how would you determine where the boundaries of individual

individuality are and that
piece i just referenced about
establishing the bounds
can be done in a few different ways one
of them would be
looking at where information theory is
um
strongest or where information flows are
occurring that's the information theory
of individuality
and here we're going to take a cognitive
perspective on
individuation and mccollum
is quoted so early in the neural network
days
the definition in the paper is i propose
a definition of an individual
based upon its information processing
structure
the scales and types of goals that a
system can pursue
defines slash determines the boundaries
and contents of the putative
agent so it's an informational lens
on what constitutes an individual like
we can say that
individual people become maybe an
individual mob
just one mob not multiple mobs when they
start
integrating information differently or
when they start pursuing
different kinds of goals in a different
way and it touches on these big
questions like
why does it matter how we define
individual
as well as what other ways of thinking

about individuality exists
yeah it's intense right so not only like
in the information theory of
individuality is it like the
the most concentrated information but
it's also like that bi-directional
information flow right it was is how uh
krakow and that group the defined
individuality in that paper and so
that's super interesting to me like at
what point
is there downward causation in an
aggregate like
so it's at at that point that is that is
like the the
the crux for how they um talked about
individuality which is i mean it makes
sense to me right like you know my
my brain tells my foot to kick the ball
or i decide i want to kick the ball and
so my body then acts accordingly but
it's um
it's really interesting to think about
when that when that hap
emerges that downward causation in a
system anyway
something i think a lot about probably
too much
what about you tara
yeah that is actually one of the few
papers i actually have read and i like
went with a highlighter and everything
um and i i always wonder on
on i was looking through the math on
that and i actually learned a lot about
like that i didn't even think was was
possible yet
but it was it was binary in a sense it

was digital
and and i'm never clear about when that
matters and when it doesn't in these
kinds of
models so that's one thing that came up
for me
sounds good next
no no it's not the sims it's
um a cone of individuation
and in 2a is um
this sort of like cone-based
representation of
individuality that's going to be
returning in a few different guises in
the paper
and on the plane going out in different
directions
from a central point is like the zone of
influence
or the extent of the system's ability to
perceive act plan and then
piercing that coming up and down in the
c dimension
is going from the past to the future and
so
this is a cool way to look at how
multi-scale systems or
what are called compound intelligences
compound intelligences
down there on the bottom right exist so
we can use this kind of a
cone representation or um to
look at single cognitive agents and
potentially contrast them
like in part b where we have a tick
with what's hypothesized to be like a
smaller scope
spatially in the now as well as a

shorter temporal scope in the past and future

compare tick with a human and then also ask maybe about what kinds of other cognitive forms exist and then these cones can be composed

really nicely so like on the example on the bottom right with an

ant colony again controversial to say that the ant is the organism

when potentially the colony is the organism that's why i usually call them an estimate

but we can see how the nest mates are inside of the colony within this cone and so the colony pushes out the boundaries

of what these ants can do so they're kind of like on different parts of the spatial plane

in the current moment and then the colony is something that is cognitively able to

move beyond the time horizon of an individual nest mate

by virtue of its collective organization and information processing

this is cool and something that we like talked about a lot like offline in preparation for this and preparing the slides is like the

inversion of the um the typical like light cone so

maybe daniel you want to walk us through this light cone and then we can

maybe go back to the other one yeah so this is a similar light cone

notion and it's just that instead of

having um
the two cups you know meeting mouth to
mouth and then
becoming pinched off in small time
another
way that light cones have been
represented is actually with the current
moment
like the eye of the needle you know like
a hourglass the pinch point
and then in the past and in the future
are the uh set of states
that influence a given point and then
the set of states it influences
so let's like think about that butterfly
flapping its wings
in the here and now so if you go back
one time
step the air molecules that might
influence it are like
one layer of air molecule away and as
you go back and back and back
more and more spatial extents matters
for the state of the butterfly in the
moment
and then similarly one time step in the
future the butterflies flapping
is only able to influence a small zone
proximal to the insect whereas further
and further out into the future
more and more spatial extents might be
modified because of the actions that
happen in the here and now
and then this is a leecors cabinet uh
you know joke on the liquor cabinet but
it's a
paper that actually uses this kind of an
approach

to do modeling of empirical data
and to fit machine learning models which
we can talk about
uh going forward like why and how
do you get a speed up of certain kinds
of simulations or data fitting
by approaching it like a light cone
so it's just interesting the axis here
like and this is kind of how i'm used to
seeing these represented are
time and space right so like in physical
space i am
physically in this space that i am at
right now
and it's interesting as like you know as
you go forward in time
like you know given enough like hours i
could theoretically like fly from new
mexico to singapore
but like for right now i mean i'm kind
of limited to like my house or my block
or you know like i don't have like
point-to-point transport
so i always like the the way that um
mike represents this light cone like in
this way like in the inverse
it's it's like time and space and and
you know it gets
it starts in a very broad space and it
proceeds to a very
narrow space and it's like of course in
the past like if i go to my distant past
like i came out of an
egg that was in my mother's uterus right
like so
or in my mother's womb like that's where
i came from i was in that one point
and i've gone all kinds of places i

guess until now but i'm not i don't feel
like i'm many places
now and and i also don't feel like i'm
going to be many places
in the future so i'm interested to to
talk to mike when he comes onto the
live stream about um the way that this
is represented
because it's very different you guys
have any more thoughts on that
yeah i have one um i mean it seems to me
he's almost trying to do
he's almost trying to take the case of
the light cone and look at a particular
ambition that you might have at this
point in time
and then and then can you carry out that
ambition through time and as you get
further and further you realize there's
all these impediments
you know that that come in your way and
so it's this like
one axis is the ability to envision
um what it is you want to do and the
other act yes this is the ability to
enact it
is the ability to well even control or
whatever your environment any number of
things that are required to enact uh
this this vision that you have
and that's an interesting interesting
access transformation
agreed um the light cone where the
uh that looks more like an hourglass is
kind of related to
our limited degrees of freedom in the
current moment
but it you know expanding out into the

future of what we can influence and then
the the diamond shape that we see in
figure two
is more like our zone of what can be
cognized
which goes beyond what we can influence
in the current moment
because there's some future time when
you won't be able to imagine
you know you know as as the the curtains
close on life
and it narrows in and it's like smaller
and smaller
and then there's some point where your
capacity to influence is actually
kind of toned out and so there are two
complementary ways to look at
the same thing although it's interesting
i want to say because
you know like you look at like there
could be some inner super meaning or
intervening thing where it actually
helps you that you never considered in
this in this cone of possibility so you
become wider like really it's just this
the winding road right
for sure
all right moving on let's go to 28. this
is sort of taking the light cone
there's just one and a half more slides
that take the light cone in a little bit
of a different
more physical direction but again it's
it's all
part of a continued conversation so we
can look at the now
that's the middle of those three layers
where there's two

different uh like individuals you know
sally and uh you know friend n
so it's like two of those are at
different points now
and so then we look over on the time
cone on the right and so agent
n is you know nick that's the one that
we're focusing on
and now the same time they're
at the same moment but they're in
different points in space
and so from the point of view of n s is
a space-like
uh point but then
in the same spot in the future or the
past
f or p in the same location you know the
same address
20 minutes ago is type like in the same
address 20 minutes in the future is also
time-like but then on the edges of this
cone is like light which i i don't know
the physics behind but i think it's kind
of a cool idea
that space and time have this
boundary nature and that there's things
that are
more space-like like things that are
instantaneously at the same moment but
in different spaces
and more time like the pure case being
the same location at a different time
but then it's like i'll meet you in 20
minutes two blocks away
that's like moving through both space
and time
and maybe light does that in a kind of
unique way

so that was just one um way that in
relativity
that they talk about light cones and
indeed the light cone concept is a
is a physicist's concept
that's really cool to think about like
that
light is the boundary between space and
time
that's kind of a neat thought yes
and then on 29 i just wanted to pull one
again
the details going beyond me
and this discussion but
here in a recent paper that explored how
you could do
simulations using light cones and so
this is not just like sort of a
metaphor or a cognitive heuristic this
is something that you can use for large
scale
data analysis and so the way which i
alluded to earlier
in which you can speed up a simulation
potentially
by using a light cone is by having a
point that you're focusing on
and then you only need to consider one
time point in the past at a radius of
one spatially
two time points in the past a radius of
two spatially
three to three out and so it helps you
reduce the space of what you need to
calculate
from all versus all to the idea of
calculating
more and more out in more and more past

steps and
more and more out as you go into the
future um
so that's because in the physical
interpretation
there's a speed of propagation through a
media
so things don't happen instantaneously
so that's like the light
moving through a media at speed of light
and then in the case of information
propagation
it's similar like if you wanted to focus
on a given person
and their role on an informational
network you would maybe make like an
informational
cone instead of needing to calculate all
by all at every single
time steps and so it just made us
curious
about how active inference related to
light cones
so how can we apply this to some of the
simulation models that we've already
seen
and then just you know what does the
past
of the light cone represent what is the
current moment what's the future
just fun eternal questions
cool ah that's so so i made this slide
this is funny this
light cone i actually showed um when i
gave a seminar
at mike levin's lab a couple years ago
it's from
um andrew gallimore's paper uh

restructuring consciousness the
psychedelic state in the light of
integrated
information theory but i was thinking
about this in terms of
shrinking and expanding the boundary of
the self which the author discusses a
lot in the paper here
um and just like how that would change
your
effective light cone right like so if
you have you have your
your self boundary it's like a a limited
like cone like the darker one that's
shown in this figure on the right
and if you expand the boundary of
yourself it expands your light cone so
it made me think of that as we were
talking as he was talking as mike was
talking about the
um shrinking and expanding the boundary
of the self in the paper
and so i'm really curious here's a
couple quotes from the paper about um
boundary establishment and maintenance
so he
says the surrounding body becomes an
informational shield
or markov blanket for the stem cell and
this is he references carl firsten's
2013 paper
um and so i wonder about like the markov
blanket
and i think that you know daniel this is
something you and i have talked about as
well
about the markov blanket like being the
defining

boundary for the individual and like how
i mean things must pop in and out of a
markov blanket
right like if i think about the cells
that are in my skin
layer right like i mean i'm losing them
all the time and new ones are constantly
being added so there's no
distinct hard line to be drawn
so like under like how do things pop in
and out of a markov blanket like
and what do you can you like just have
like peripheral
boundary nodes on the edges there like
i've talked before also about like the
difference between like your fingernail
and your cuticle
like there's like a a small tiny layer
that doesn't know whether it's a
fingernail or a cuticle
so it's like how do you like have these
like boundaries
right um anyway another quote from the
paper
crucially by organizing into a partial
electrochemical
and informational pool or sync to
tm sinks somebody say that for me please
syncidium syncidium there we go
all of the cells are able to measure and
detect events occurring
within the same boundary creating a
larger individual
that emerges from the collective and i
think this quote was like in
in reference to the gap junctions and
how they expand that electrochemical
pool by like

they open up like some like they pull
the plug on the electrochemical gradient
and so it
kind of enables this like cell linkage
so that's one
one way that um you know
boundaries are maybe established but
any thoughts on this guys
we'll come back to it i think i have a
few thoughts on where the markov
blankets come into play through time
cool all right
uh so homeostasis which we we promised
to come back to so
what evolutionary pressures led
cognitive boundaries to the
to expand this is from the paper and the
author proposes that the atom
of this cognitive hierarchy is
homeostasis
um so reactive homeostasis evolves into
predictive allostasis
under the pressure to predict signals
from the environment and
other elements of the biosphere and
that's like what i was talking about
earlier about
the first like counterfactual about um
you know what if it's not always 76
degrees
and full of nutrients in my environment
how can i regulate that reduce my
uncertainty about future availability of
the supplies that i need
and also there's this um nice
progression
from homeostasis so that's sort of like
similar

you know same homeostasis it's the thermometer is trying to get within the bound but the best way to stay within the bounds when the environment is changing is actually to not just be reactive but to be proactive or predictive and then what's the best way to be

proactive well it's to have increasingly rich

generative models of the niche if you know that night time is coming you're going to be able to take really good predictive behavior right now for tonight if you know that winter is coming at a larger time scale it's another level of you being able to make adaptive decisions

so what looks like homeostasis sometimes can actually just be very uh skillful predictive allostasis

[Music]

do you have any thoughts sarah on that cool um

okay so this one is memory and scale free cognition and active inference um so

i don't know i just had some thoughts here like memory

in as a gene regulatory network and this is kind of something that not really was stated in this paper but really like

you know it's it's our evolutionary memory of like where we've been before and how we've um come to

regulate and you know work within our
environmental condition
and like also memory is a process like i
don't know
i think about that like where is it
stored is it externalized into the niche
like we do a lot of niche modification
and so
i don't know um it's interesting to
think about
our our ram and our rom in um
in like a biological sense so
it's the author says a second step in
the simplest homeostatic loop is the
inclusion of a richer set of hidden
layers in the neural network sense
so these are additional biochemical
nodes between sensors and effectors of a
given system that enable a degree of
memory
and importantly memory can serve as the
beginning of modularity
because learning essentially groups
diverse stimuli into compressed
representations
complex states of affairs become
remembered as compact
biophysical engrams this is the essence
of the kind of modularity when a simple
biophysical event kicks off the
formation of a complex
morphogenetic cascades such as building
a hand in embryogenesis
one nice point there is the difference
between modularity
and memory about how they enable each
other
it's sort of like if everything is being

mixed in one
cup there can't be a memory of different
colors okay well if you have two
separated cups
you can remember two different colors
and so on
so memory can serve as the beginning of
modularity
is saying that it's this um relationship
between things becoming distinct
and being remembered through time
distinctness carried through time
is memory you can remember a longer
number
when you have more modularity that's
being propagated through time
and then the paper on the right is um a
an
older first in paper talking about
memory attention and salience and active
inference
where they described uh memory working
memory as
a process of evidence accumulation in a
temporarily structured hierarchy
so i guess like going back through deep
time like that's where the gene
regulatory network kind of comes in
as a process right like the process of
evolution
as and that's like the working memory
there anyway
and just you have one one note there on
the multi-scale active inference models
sometimes the way that memory is being
modeled inactive inference
is as a state variable that changes at a
slower

rate and so within the longer
time scale model we have a shorter time
scale model
and then that can be modeled from
setting to setting
as memory carrying over and so it
relates to this idea that
the larger cognitive system is able to
also act over longer time scales as well
as reach back
with a deeper memory because if you have
the kind of
short-term model that's just stand-alone
it doesn't it's like a memory-less
process but if that has inherited
from a hierarchical model the state from
a previous
um click of the model it's memory
and then if it's able to imagine one
more click out into the future
it's goal planning it's anticipation and
so inactive inference
one of the ways that we integrate memory
and planning together is by thinking
about
hierarchically nested models so also i
wonder what role memory plays in
policy selection like i remember every
summer
when i go to scotland it is
cold like i mean i live in new mexico
it's like 120 degrees here
fahrenheit so every time every summer i
remember that when i go to scotland it's
cold so now it's my policy
to bring a jacket when i go right so
even though it's the summer time and i
wouldn't normally do that so

is that memory also ever modeled in the
policy selection daniel

it's a good question like i wonder if
you can

plan to forget or you know i'll make a
note i'll plan to remember

i'd say active inference just gives us a
way to

think about all those different
affordances that real cognitive systems
have like

offloading memory to the niche through
external computation or in terms of what
you

just described but that's a good
question we can ask

right i don't actually even need to
remember that it's cold in scotland
because you know i can just ask google
they will tell me the weather and it
will be cold in scotland so if i just
remember to

you know check my cognitive offloading
device

then i don't even need to remember to
break a jacket when i go so

yep okay

next so i figure i have one i have one
random thought maybe it comes up in
another slide but is this question that
michael levin has about how does it know
when to stop in terms of um

cell growth and morphogenesis if it's
the case that it's connected to memory
in the way that he's suggesting then
yeah you know like maybe if it related
the information flow

i guess that may be obvious given the

focus of this

um but it is interesting you know it's

like

the the cells are just like we've got

we've got to

now become multi-body you know

or become social rather than individual

in order to store memory like there's a

funny like

break point yeah i didn't see the stop

mechanism really brought up

in this paper but i have seen it so

um i'm not really sure where it ties in

here

um figure three

i mean i mean if he calls it a

morphogenic cascade you know it's like

well we're

we we we're born we grow we grow to a

certain point and then we shrivel and

die

so it's like how does that connect to

memory and information

ah it's kind of suggested

yep so figure three

is looking at a sort of cell's eye

view of that time cone

and now we're going to step away from

you know is it a time diamond or is the

time going

and we're going to think about the

trajectory of a given cell

go back to the assumptions of the paper

though this isn't about whether the cell

is experiencing things it's not

what it's like to be a cell it's about

how

the cell has a given spatial extent in a

it has a given spatial extent that can
be thought of as having a memory
it inherits a history through time and
then also has
an anticipation zone in the future
then it can do anticipation of memory
as a sole agent or it can interact
socially
so what happens when it interacts
socially with other agents
like itself well it expands the spatial
dimension in the now that's what's being
called
integrated spatial perception and that
allows it to
have expanded memory in the past like if
each of your
five friends have a book then there's
more memory as a group because there's
more books
and then also the ability to extend the
anticipation
into the future and then this part i
just highlighted in blue
is joining into communication networks
allows the cell to have
temporally delayed access to the
information obtained by neighboring
cells
this is where the time cone comes into
play it's not that when something is
perceived
um by one of these cells off on the
fringe
the vocal cell doesn't instantly hear
about it
they hear about it one time step later
so that's where the times the time cone

comes into play
because from the point of view of this
focal cell
just assuming that um information moves
one cell per unit of time
then at zero time points its
radius is one one time point away
its ability to integrate sense is one
more radius
away and so you can keep on clicking
back and that's reflecting a broader and
broader zone of memory
or you can carry it forward with a
broader broader zone of influences or
states that it can affect and so the
reason the resulting
larger individual and again individual
is being used cognitively here not like
evolutionarily is formed with a
cognitive world that's unified
by the cells sharing of information
across time and space
so no teleportation and no time
travel the limitations of propagation of
information
in our physical universe means that from
the point of any given cell
or agent it's existing in this time
cone-like structure
yeah that was like the expanding the
boundary of the self um
and like kind of like what happens like
the time cone is expanded and you know
what this just makes me think of well
first
i just want to remind everyone that this
like cell cell communication
is like what is what gap junctions

facilitate because they're
really connecting all cells and solid
tissues um but but really i thought
about
you know the old adage that you are your
company and like if you think about you
know your five like closest friends
and that like communication like network
that you have like you
hear about you know world events or
events that
are relevant to their lives it might be
relevant to you like you hear about it
one time step later and just how like
you know your cognitive
like boundary is expanded by like your
inner circle
i don't know made me think of that
yep hey blue can you just can you just
re-highlight or
talk about the cell uh cell exemption
that you're just mentioning
oh yeah so that was earlier in the talk
it was one of the key words so let me
just go back to that slide
um so here uh these gap junctions
so they they are they were originally
you know categorized
in like neural tissue and uh in muscle
you know
to facilitate like rapid communication
but they're really present in every cell
in solid tissue and they they just
perform um not ion exchange but it's
like passive diffusion of ions
so that it's like almost a shared
intercellular space
so would it be

safe to call that a kind of a
information
bottleneck i just wonder so maybe not as
much of a bottleneck as it is like
um it pokes a hole in the informational
boundary
yeah okay so that it's more the more
rapidly
let's just say that the gap junction so
the no gap junction
it moves like um takes three time steps
for the signal you know one time step to
release the molecule one for it to
diffuse
one for it to be perceived so every
three units a time
you're getting a radius expanding out by
one now let's just say that a gap
junction allows
one time step the direct diffusion so
then
the time cone is expanding wider
because the diffusion from the target
cell is able to sweep
out a larger time cone whereas if again
if it took three units of time for every
one of diffusion it'd be like a narrower
time cup
so rap communication which is never
instant
allows that time cone to access more
memory
into the present and then influence
more broadly and so bottleneck is sort
of like thinking like a bottleneck is a
constraint
relative to the width of the bottle but
in the

context of a gap junction it's actually
like a channel or an opening for
information transmission
so what do you think i'm just i'm just
gonna go ahead
if you think about it in terms of like
your um inner circle of friends
imagine like you can wait for your
friend to call you up and tell you this
exciting new event or like you can have
a wire tap
into their house so like the gap
junction is like having a wire tap into
their house so like you always know like
what is going on
right okay
i'm also searching for another analogy
that's just saying into the stuff i've
been looking at lately do you think it's
fair that
analysis to um you know a compression
function
like you know if you picture the classic
neural net you know a bunch of nodes
here just a few in the center of the
hidden layer and then expanded out would
would it better be a fair analogy for a
gap junction
yeah i'm not seeing it sorry okay
okay i would just uh to recover it i
would just say
that the transmission is like a
compressed form
of it's like a little sampling of the
chemical milieu from one cell being
passed to the other
if there was no bottleneck no
compression of information

they would be just in one unit there'd
be in one markov blanket
and so there's this trade-off where if
you totally isolate
the rate of information propagation is
low the modularity is very high though
on the other hand with total integration
you can't have memory
because you don't have modularity
however you do have
instantaneous or close to that
transmission
and then the gap junction exists at this
kind of sweet spot
that allows for modularity with
narrow channels of communication which
is what
shannon was studying in the first place
cool so figure okay so figure three i
think we just put popped on the cone is
that true
yeah that's just to draw the connection
between the figure two with the
the time diamond time cone and then just
this
figure cool okay
figure four uh so
figure three oh yeah well the the figure
three the
slide that we just had was like as
things build out so that was the point
of view of the agent building out and
now it's like things fall apart
so figure four we see a sort of
rehearsal
in b of what we saw previously which is
that that target cell
it's combining sensory information

that's able
enabling it to detect patterns more
rapidly getting a big picture
that's what it says in b and therefore
missing information can be anticipated
by this pattern
seeking drive but what happens when
things
go wrong that's in c where this bottom
right
cell the green one has gone rogue
it has stopped communicating in a way
that was participating in this broader
sense making
and it has shrunk its sensory radius
down
and so in doing so it reduced the
cognitive capacity
of the other four cells it has reduced
its own sphere of influence
and it's also introduced this
adversarial dynamic potentially
over evolutionary or developmental time
skills
cool any more thoughts on that one
reminds me of a friend i know but no
other way
it also reminds me about how cancer
research is starting to think more about
the niche
and the signaling between cancer and
non-cancer cells like cancer cells need
to call down
healthy blood vessels they need to
prevent
healthy immune response from taking them
out and so
it's like cancer as a dialectic or as a

relationship

between agents with different goals

again not the experience

of the goal but acting as if it has a

goal that third person cybernetic

teleology

and so that helps us make sense of all

these disparate features

of cancer by thinking about internal and

external states

information propagation etc what do you

think it would be right or off to think

about the

one possible place where that could go

wrong

um happening at the gap junction like

the gap junction would no longer

function as a good sampler compression

thing which i know

you don't agree with the analogy but

something like that i

like i i don't know what's happening in

that research area but

does that make any sense yeah we talked

about gap junctions earlier and their

involvement

in um neoplasticity and all kinds of

other disorders

definitely i just found this really

nice paper called gap junctions in

cancer

communicating for 50 years and it's the

abstract starts with

50 years ago tumor cells were found to

lack electrical coupling

leading to the hypothesis that loss of

direct intracellular communication

is commonly associated with cancer

progression and onset
so that's pretty interesting it's like
when you
shut down the lines of communication
like between two countries
are those two countries having an
improving relationship or is that like
the last time that they spoke before war
so closing down the embassy
shutting down the gap junction it
doesn't bode well
for a sort of amicable resolution to
conflict or communication
that's why it's so important to maintain
the avenues of communication
even when there is an adversarial
relationship because things can always
get worse
you can always shut down the
communication which will make the
generative models and the incentives and
the goals
diverge even more
cool i don't know just all of a sudden
you go into politics and i'm like oh
yeah but wait a minute
anyway no but but we should shut down
they're bad we should show
if them keep the channel open you also
stand more chance of injury
by way of keeping the channel open so
keep the channel open until you
shouldn't um
this is just um a fun uh discussion for
a little bit
on the evolutionary anachronism which
means like
uh just something that's happening at a

different time
than it's expected like a feature from
the past that gets brought back
like an evolutionary reversion
and so in the paper it's described
that uh this is a breakdown of
multicellularity and highlights the fact
that the scale of the structure which
cells work to maintain can change
rapidly
cooperation towards an entire organ
system or body
that's a healthy organism to the level
of a single cell
so instead of seeing cancer just
mechanistically as a derangement of
signaling or derangement
of replication and dna reproduce
reproduction of a cell
we can think about it informationally
and cognitively
and cybernetically like the cell has
changed its cybernetic horizon
from the goals of the organism to its
own local goals
and so this concept of an evolutionary
anachronism and how
even in the current day we can study
anachronism
uh in examples with avocados a tasty
fruit
and as per the wikipedia it says
avocados are exceptionally fatty fruits
with seeds far too large to be
successfully dispersed
by any wild animal presently alive in
the americas
and so that is a paper of jansen and

martin from the 80s
which was actually talking about the
ecology
of large animal plant interactions
of animals that we've never directly
seen but it's like if you
know that there is a really large seed
and you know that there's seed
dispersing mammals
then it can lead you to hypothesize that
there was a seed disperser that was
large enough to eat
avocados whole um and
so we can look at anachronism in the
current moments
and help us you know retrace the past
so just like the avocado is an
anachronistic
trait with a such large seed given to
the current ecology
past ecology we can see cancer as this
anachronistic
reversion to a previous mode of
information integration
and goal setting that it's like it's
inappropriate
in the context of the goals of the body
but
not necessarily within the goals of a
single cell that just wakes up one day
it's like wow i'm bathed in glucose
i should replicate so can't blame that
stuff
also like the cancer a lot of the
markers in cancer cells are
really like it's like a developmental
reversion
so not sure if you guys are aware of

that or not but it's like they they
revert to like a more immature cellular
phenotype

and so that i mean it's literally like
going back to the past

so it's interesting

[Music]

like is there something can be learned
from the degree of reversion

uh the degree of

i'm not sure here's one thing that comes
to mind there is there's

some cancers where the the it
goes back to like a stem cell of that
tissue

so like it's like a white blood cell

cancer that like

replicates a lot of white blood cells

but

when there is a cancer in a reproductive
organ

sometimes it's like called a teratoma

because it can have

like hair and nails and stuff like that

because it's actually pulled back so far

that it can then

differentiate on all the pathways

rather than a reversion up from being

like a

healthy reproducing white blood cell

precursor to just a

higher like a white blood cell stem

lineage but it doesn't pull back all the

way to including ectoderm

whereas if it was in um you know a

reproductive cell then it could do that

cool biofilms

do you want to take this one danielle

yep so i think the
key point here is that bacteria have ion
channels
just like metazoans do you know dopamine
gpcrs the proteins that signal between
neurons
those are not innovations of mammals
they're not innovations of vertebrates
they're those kinds of channels and
diffusible molecules
are cellular they're pan cellular
and so what's being shown here is just
like in
previous figures we saw one kind of
eukaryotic cell right
you knew because it had a nucleus we had
one eukaryote
connecting with a few other eukaryotes
well here
we have one prokaryote connecting with
other prokaryotes
to do the exact same thing which is
increase its
sensory capacity again not
instantaneously
but think time cone like cone
and so in the same way that the
eukaryotic cells form a cognitive
individual
through their associations and
affordances such as the gap junctions
bacteria when they form a biofilm can
also form a larger cognitive integrated
individual
and you can still say well but from an
evolutionary point of view
this the unit of selection is the single
bacteria okay

that's the evolutionary individuality
definition
and when we use the cognitive
individuality definition we see a lot of
the similarities between
a multicellular organisms tissue and
then a multicellular
biofilm tissue even not in a biofilm i
mean bacteria have
an incredible ability to like i mean
it's like antibiotic resistance right
like they do plasmid exchange which
isn't like their chromosomal dna but
it's like small
little snips of dna like they just like
have a handshake like it's like a
bacterial handshake like now
i'm gonna give you some antibiotic
resistance too like we spread
germs like covet and bacteria to one
another they spread
like plasmids to one another which which
confer different properties you know and
then
we select for this like antibiotic
resistance through the use of all of our
like antibacterials and so only bacteria
that have these little plasmids are
survive it's really interesting
so even outside of a biofilm they do
some some kind of um
you know communication and like
cognitive expansion of their self in
like the way that we do with our inner
circle of friends
cool all right key ideas um i just
highlighted a few of these
like points but there's a huge box in

the paper so we'll just read through the highlighted parts

here i'll take this one so first

a unified integrated cognitive self or individual can be defined with respect to the integrated ability to pursue specific goals via a homeostatic process that resists perturbations

second an agent's cognitive world can be quantified and characterized enabling comparison with others regardless of their material implementation

by estimating the spatio-temporal boundaries of its area of concern the volume in space and time over which the agent is able to take measurements exert influence

and functionally link disparate events learning and association

three the borders of the temporal and spatial events of which a given system is

capable of measuring and acting map out a cognitive light cone

a boundary in the informational space of a mind

so estimating the spatio temporal boundaries of an agent is something i find particularly interesting so i just

put that in as a question at the bottom do you want to read some of the next ones daniel

yep so these are some key ideas four five six

so four cancer is a reversible shrinking of the computational boundary of a biosystem

pretty tantalizing there with the
reversible i'm sure
that's something that a lot of people
would like to see maybe
the idea is you remind the rogue agents
about bigger goal like hey you know i
know you're hungry
but we wait in line at this restaurant
so that everybody
gets what takes care of them and maybe
that helps remove the you know
cancerous uh restaurateur five
agents scaled up by evolving from basic
homeostatic loops
driven by active inference which is
surprise minimization
by the addition of delays which are
memory anticipation which is inference
inference has planning as inference and
networks
which is spatially distributed
processing enabling learning
and progressive abstraction
generalization and data that's pretty
cool
to see active inference and elaborations
of active immigrants
at the heart of this future electrical
evo eco diva it's like saying
you start with action and perception and
then you can
carry that forward in time planning
anticipation
have an extended trail of memory that's
you know the past and then you can also
expand spatially but again a spatial
isn't instantaneous
there's a propagation speed in the media

and so even laterally
similar agents interact through this
light cone way
so nice to see where active inference
plays into like cones there
and then six infotaxis the drive for
better actionable intelligence
about the regularities and patterns in
the world and in the agent's own
mechanism
encourages cells to connect in groups
via signaling
this is another area where active
inference has a lot to say
and is a value add relative to a reward
only framework like reinforcement
learning
because again what drives the bacteria
to coalesce and to work together
so maybe it's reward but then you have
to appeal to this idea
that they're maximizing reward with
their behavior and the state or leave
and all this kind of stuff
info taxes helps us understand that the
way to get
good actionable intelligence which is
kind of like active
inference a i right there is to make
your actions
exist on the trade-off frontier like the
pareto optimal trade-off
between getting it done right now and
expanding your epistemic boundaries
sometimes the right move is to contact
somebody who knows better
or to go to a website that might inform
you better and so it's actually a useful

action

in the moments to arrange to get better
information later

and so active inference helps us think
about how these different needs are
traded off by

organisms cool

all right so seven collecting into oh i
can't do it

syncidium sensitium do i do it

okay collecting into a

syncidium enables all of the cells to
share the same data

and access the same memories

um that's pretty awesome you must
assimilate it's like the borg

yep sorry star wars nerd um okay also

greed eight greed at the single cell
level for information

infotaxis drives cooperativity as each
unit

expands its measurement boundary

communication with neighbors

and thus inevitably becomes part of a
bigger self with bigger

set points serving as homeostatic
attractors

so i think active inference also has a
role to play here

in this cooperativity so like i can
reduce my uncertainty about my
environment

by cooperating with people who have the
same

goal that i have right like or
cooperating

becoming part of a group you know and
and therefore our efforts

make a bigger effort or bigger
anyway less surprise when i'm in a
bigger group because it
expands that that light cone right and
then nine
there is a fundamental symmetry between
anatomical control mechanisms
and cognitive mechanisms and
um i think that this was discussed in
terms of bioelectricity
in the paper do you guys have anything
to add
one thought on that is cooperation
arising at a higher level
so that just reminds me of like if
you're looking for a microscope and you
saw a cell a eat cell b
you'd go i mean adversarial relationship
right could there be anything more
directly adversarial
and then you pull back and it's a white
blood cell
you know that's attacking a cancer cell
so it's cooperative at a higher
level but it can look adversarial at a
low
level and so what does that
map onto in different systems how can we
re-imagine conflict and tension as
actually playing a stabilizing or
cooperative
role at a larger scale can interpersonal
differences in perspective or
disagreements lead to a cooperative or
regenerative collective this is what we
hopefully move towards
i saw some pigeons procreating yesterday
while i was out walking around in my

neighborhood and it definitely looked
adversarial like it looked very
very adversarial but they were
cooperating at a higher level
looks can be deceiving uh okay
definitely i was thinking of a case
where like if a tiger attacks me i
should just assume that it's like
an information transfer that needs to
happen i don't know
come get peace with it
sarah oh it's like in um the selfish
gene by
dawkins he says uh yeah why would you go
beyond the gene
does it make any sense you know would
you have mammals like uh
doing kind acts for each other just look
at the lion eating the gazelle
and how could that possibly be a
collaborative relationship and then i
thought
what if those two populations stabilize
each other through deep time
and they can't coexist and then they
actually
persist better both because of that
one act so who's to say
but just one thing like if that is
all the case you know if this is all
like whatever man the universe becoming
like why would it be that that we had
such a hard time with it like
shouldn't it shouldn't there be some
mechanism that allows me to just be like
oh okay
cool you know like i don't get that
i think the mechanism is narrative

cognitively

it might play out differently in

different systems

but agree to disagree it's a narrative

that we can both

agree on we're cooperating at that

narrative level doesn't mean that we

don't

disagree in fact we entrench that we

disagree

but we're agreeing at a higher level

cool do you want to read the next ones

so yeah just a few of the key points

from 10 11 12.

uh 10 bioelectric integration helped

evolve control strategies and cognitive

content

across the continuum from chemical

networks to human minds

11 the key points here were that there's

a scale

in variance with how different systems

make decisions

and that we can across different systems

study how the cybernetic

processes of learning and parameter

optimization

are implemented by a large number of

units pursuing

infotaxes and homeostatic goals aka

pragmatic value and episodic value and

12. a conceptual unification is proposed

in this paper

as scale-free cognition and um

then it's talked about how these

boundaries between different subunits

are a major control mob between the self

and the world

like the cell is doing the same stuff
whether it's part of an organism or a
part of a cancer
but it's actually this control knob like
what are the limits of the cognitive
individual
that sets the cell up to act a certain
way or another
and uh then a reminder at the very end
from a utilitarian perspective that the
most appropriate level of analysis
is to be determined empirically aha was
empiricism
determined empirically and that is said
to be
the level that facilitates prediction
and control
with the least effort by the
experimenter or by the system itself
what if they disagree and gives rise to
a unified understanding that drives the
most novel
robust research programs at the bench
nice
you know if it doesn't advance your
career as a scientist
that's not the case
this is you daniel active inference in
multi-scale communication
just wanted to capture a little bit
about multi-scale systems and active
so here we have nested systems so the
top left you know that's an ant brain
that's a picture that sasha took of an
ant brain that came from a pogo ant
and then there's the colony with a bunch
of nest mates
dragging food around and then the colony

is embedded within an ecosystem
so within one nested system you know
within these
time diamonds that are sort of nested
inside of each other
we can think about how the internal
state of one
is linked to the external state being
the next level up
like the internal state is the brain and
the external niche to the brain is the
body okay now the body and the brain are
the internal state
for what the colony and then the colony
is like the internal state
for the ecological niche so that's
something that we've seen before
with how active inference helps us
understand and model and simulate
hierarchically nested systems but also
we can
think about interactions at the same
scale like collective behavior
collective intelligence as occurring
when
at the same of agent they're interacting
like laterally and so that's this teal
arrow
that's where you have two agents there
are they're each other's external state
so active inference helps us zoom up and
out
and down and in and then it also helps
us go laterally
and then not to repeat it too many times
but the lateral
transmission isn't instantaneous it's
dynamic it happens through time

which means that we don't need to look
at all

lateral interactions just the time cone
cool uh predictions and research program

uh there was a big list here so i just
pulled a few

of them out i'll read some of these off
which are a lot of them questions so is
there a quantifiable sense
in which biological systems model
themselves

can an in silico evolutionary system be
built containing both genetic
and physiological components which
simulates the scheme described above
homeostasis and infotaxis and

illustrates the emergence of different
scales of cognitive horizons over time
can we directly observe the evolution of
multicellular goals from networking of
agents

with single cell homeostatic goals
might there be higher level effects on
the physiological boundaries
of the cognitive sub-agents within the
body from exposure to psychedelics which
have long been claimed to expand
transpersonal boundaries or induce ego
death

the analogies between geometric spaces
and cognitive ones need to be explored
further as

has begun beginning to be done via
geometric

information theory time diamond anyone
definitely i think that there's a lot a
lot of room to go here

will complex engineered systems set

novel goals
and if so how it is a prediction of this
view
that adaptive useful robotics will
require components that are themselves
competent and
goal seeking that is super interesting
also um because really as an intelligent
you know being capable of cognition so
are all of my parts
so if we're going to really import you
know give rise to
intelligent life intel intelligent
artificial life
we have to start with intelligent
artificial components i think that's
super cool
and also goals we can't we can't just
say well let's just build
you know a bigger car or a bigger
computer i mean what
do we want this intelligence design
system to be intelligent
in the service of or in the way of
and uh it's a disaster waiting to build
capacity without goal
for sure do you want to read some more
of these
yep so the second set of predictions so
this is where
mike is setting out a little bit of his
uh agenda the next work in this
subfield i just you know the subfield
each of the fields
they're modular they have their own
disciplinary histories their own
disciplinary anticipation but this
subfield

should focus on the discovery of
bioelectric communication inducing
technology
so just like once we had chemical
biology people were really excited about
finding drugs that bound
to proteins well now that we're aware of
bioelectricity let's think about
manipulations and measurements of
bioelectricity
two it's a prediction that it's possible
to induce the formation of metazoan like
bodies in an otherwise unicellular
organism
by forcing the expression of appropriate
electrogenic proteins and gap junctions
so he's trying to get towards some
unique predictions and experiments
uh prediction that transcriptomic which
means gene expression
analyses of regeneration events from
bacterial biofilms plants and metazoan
models of limb regeneration
should show consistent use of
electrogenic proteins
in the events that enable cooperation
towards self-limiting
morphogenetic cascades so it's kind of
like
the um the structure informational or
cognitive structure
of recovery and repair might have some
similarities across systems
that wouldn't immediately be seen as
similar
otherwise scale-free cognition
hypothesis
suggests that multihuman systems could

remote teams could have their own
degree of cognition so as we always say
you know things that the brain knows
that the neuron doesn't things that the
colony knows that the ants decimate
doesn't

there's got to be things that remote
teams or any other kind of team
know that individual participants don't
know

and then the exobiology question is
given a novel

life form or just a novel form how does
one know when successful

communication has taken place and
that's an interesting question for a lot
of you zeno slash

exobiologist seti anyone

cool next slide so this is in the
section

what it feels like to be a pancreas but
i i really pulled um

the point that kind of hit home for me
the most out of the section

and the final point that mike makes in
the paper

so i'll just read this off it is

striking that the process

which zen practice is meant to reverse
attachment to past memories and high
valence for future

expectations or fears is precisely the
process

suggested to be responsible for the
creation of complex

selves with a capital s it is unclear

whether it is beneficial or even

possible to truly live in the moment and

let go of past memories and future
expectations

but anyone who succeeded in doing this
would achieve precisely what zen
promises

the dissolution of the self and the
final sentence of the paper which i just
loved

recreating the unification into and
liberation

from larger scale unifying unifying
cells with a capital s

would be a true pinnacle of synthetic
biology and artificial

intelligence engineering wow

this is good paper

um so just key questions at the end what
might a good understanding of this work
enable

what are the unique predictions and
implications of this work

what are the next steps for fep and
active inference

in relation to this work what are the
goals of this research and what are you
still

curious about so we will be thinking of
all of these things

as we get ready to do um 25.1 and 25.2
over the next couple weeks

so thank you for participating there is
a feedback form for feedback

and you can get in touch with active
inference lab at

all of these links on this slide

nice work blue great job with your first
broadcasting and facilitating on this
one that's really

fun so yeah anyone's welcome to
participate or contribute to our live
activities
asynchronously synchronously we'll
figure out the right performance for you
so thanks sir also for joining thanks
again blue
cool